Project Proposal: Clinical Stroke Risk Classification

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Dataset: Stroke Prediction Dataset

1. **Objective & Problem Scope**

Stroke causes ~11% of deaths globally. The goal is to build an end-to-end binary classification pipeline that predicts stroke risk (stroke = 1) from physiological and demographic indicators to support proactive clinical intervention.

2. Dataset Overview

- Scale: 5,110 rows, 12 attributes (meets course minimum of 2,000 rows).
- Target: stroke (1 = stroke, 0 = no stroke).
- Demographic / Lifestyle: gender, age, ever_married, work_type, Residence_type, smoking_status.
- Clinical Features: hypertension, heart_disease, avg_glucose_level, bmi.
- Identifier: id (to be removed).

3. Preprocessing & Leakage-Free Architecture

All preprocessing parameters are learned strictly on the 70% training split before transforming the 30% holdout test set:

- Train-Test Split: Stratified 70/30 split executed first to lock holdout distributions.
- Cleaning & Pruning: Drop id (non-predictive key) and drop the single 'Other' entry in gender.
- Missing Value Imputation: Impute 201 missing values in bmi (~3.9%) using the training set median BMI.
- Categorical Encoding: Apply One-Hot Encoding (drop_first=True) to gender, ever_married, work_type, Residence_type, and smoking_status.
- Feature Scaling: Fit StandardScaler strictly on training data for age, avg_glucose_level, and bmi.
- Class Imbalance Mitigation: Address the ~19:1 class disparity (249 positives vs. 4,861 negatives) by fitting SMOTE exclusively on the training fold.

4. Evaluation Strategy

Because raw accuracy is distorted by the 95% majority class, performance will be benchmarked using:

- Primary Metric: Recall (Sensitivity) to minimize dangerous false negatives.
- Threshold Metrics: Precision-Recall AUC (PR-AUC) and ROC-AUC.
- Baseline Classifiers: Logistic Regression, Random Forest, and Gradient Boosting.